TECHNICAL WHITE PAPER

Public Health Signal Detection and Response Dashboard

Applied Public Health Informatics Engineering Prototype for Surveillance Analytics Modernization

Document Attribute	Description
Purpose	Technical white paper for surveillance analytics modernization and applied public health AI engineering.
System scope	Syndromic surveillance ETL, anomaly detection, forecasting, geospatial risk scoring, dashboard intelligence, RAG-assisted query, and synthetic HL7 simulation concepts.
Data boundary	Public aggregate and synthetic surveillance data; 50,000 raw rows transformed into 7,842 processed analytic records.
Prepared date	May 06, 2026

Executive Summary

Syndromic surveillance is a foundational public health intelligence capability because it allows agencies to monitor illness patterns before laboratory confirmation, case investigation, or final diagnosis coding is available. Emergency department and urgent care encounter patterns can provide early operational signals for respiratory illness, gastrointestinal outbreaks, overdose-related harms, heat illness, environmental exposures, and emergency events. At national scale, these signals support preparedness planning, cross-jurisdiction awareness, and faster allocation of analytic attention during uncertain public health conditions.

The Public Health Signal Detection and Response Dashboard is an applied public health informatics engineering prototype designed to demonstrate how modern analytics architecture can convert public or synthetic syndromic surveillance data into interpretable operational intelligence. The system integrates ETL, quality control, temporal feature engineering, rolling baselines, statistical anomaly detection, Isolation Forest scoring, risk stratification, geospatial ranking, forecasting, and analyst-facing visualizations. It also includes a retrieval-augmented natural language assistant pattern for grounded query and briefing workflows.

The platform is intentionally designed around explainable model outputs rather than opaque automation. Each signal is accompanied by baseline context, observed activity, anomaly scores, percent

deviation, risk level, and review-oriented action logic. The architecture reflects healthcare data modernization priorities: modular pipelines, interoperable data concepts, governance-aware analytics, traceable model outputs, and a future path toward cloud-native streaming ingestion, standards-based HL7/FHIR integration, and controlled retrieval over surveillance knowledge assets.

Processed records	Raw source rows	States/regions	Syndromes	Anomaly flags	Completeness
7,842	50,000	51	3	236	98.2%

Table 1. Current prototype metrics computed from the repository processed surveillance dataset.

Public Health Context and Surveillance Modernization Need

Modern public health surveillance requires more than retrospective reporting. Agencies need timely, interoperable, and analytically usable data flows that can support situational awareness during routine monitoring and emergency response. Syndromic surveillance addresses this need by monitoring symptoms, chief complaints, discharge diagnoses, and visit patterns before confirmed diagnoses are available. This approach is especially relevant for early outbreak detection, respiratory illness monitoring, opioid and overdose surveillance, environmental health events, and emergency preparedness intelligence.

The prototype is informed by concepts used in CDC NSSP, ESSENCE, the BioSense Platform, HHS public health data modernization priorities, and WHO epidemic intelligence practices. These references establish the operational context: surveillance systems must ingest heterogeneous healthcare data, standardize it into usable analytic representations, detect unusual patterns, support cross-jurisdiction interpretation, and communicate findings to analysts without overstating automated conclusions.

Reference Domain	Operational Relevance to the Prototype
CDC NSSP / ESSENCE concepts	Near real-time syndromic surveillance, analytic queries, visualization, trend monitoring, and signal investigation workflows.
BioSense Platform concepts	Cloud-oriented public health data processing and monitoring infrastructure supporting early detection and situational awareness.
HHS modernization priorities	Timely, interoperable, governed data assets that improve preparedness, response, and public health decision support.
WHO epidemic intelligence	Detection, triage, verification, and assessment of signals from multiple sources to support public health risk awareness.
Emergency preparedness use cases	Respiratory surges, overdose-related events, heat illness, disaster response, environmental exposures, and cross-jurisdiction monitoring.

System Architecture

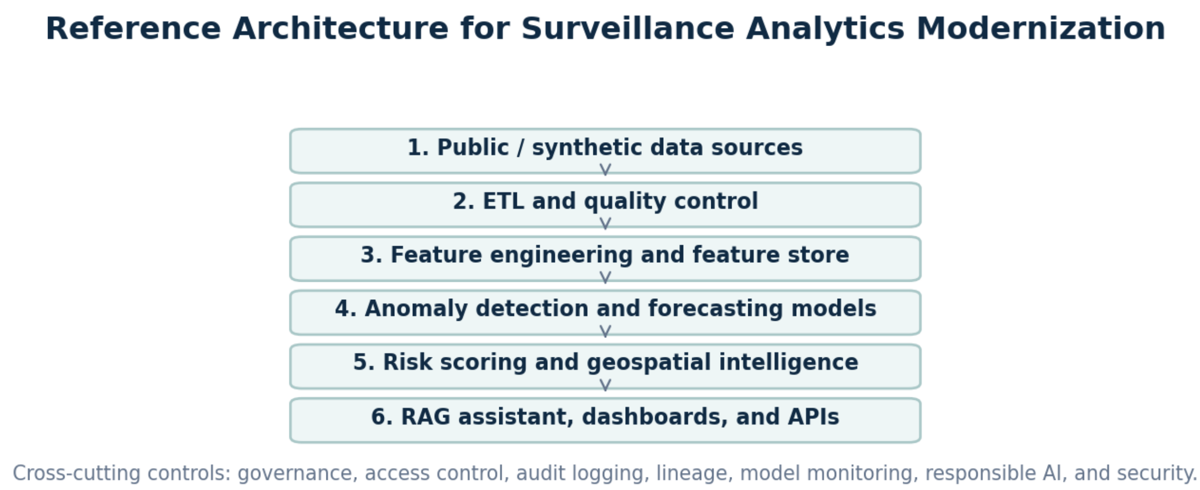

Figure 1. Reference architecture for modular syndromic surveillance analytics, including governance controls across the full data and model lifecycle.

The architecture is organized as a modular analytics pipeline rather than a monolithic dashboard script. The design separates ingestion, cleaning, feature engineering, model scoring, risk interpretation, geospatial visualization, retrieval, and presentation. This separation supports testability, governance, model replacement, cloud migration, and future API exposure.

Architecture Layer	Engineering Role
Data ingestion layer	Accepts local public/synthetic datasets today; can be extended to scheduled batch jobs, secure object storage, API feeds, HL7 messages, FHIR resources, and event streams.
ETL and quality layer	Normalizes dates, geography, syndrome labels, numeric values, missingness, duplicates, and analytic grain before model execution.
Feature engineering layer	Computes lag features, rolling baselines, rolling standard deviations, z-scores, percent change, and temporal acceleration by state-syndrome series.
Analytics engine	Executes statistical anomaly detection, Isolation Forest scoring, model agreement logic, forecasting, and risk stratification.
Geospatial engine	Aggregates risk by geography and supports state/region ranking, heat-map views, and spatial prioritization.
RAG/NLP layer	Retrieves grounded surveillance summaries, model outputs, quality reports, methodology notes, and reference guidance for analyst-facing responses.
Presentation/API layer	Provides Streamlit dashboard views today; future architecture can expose REST APIs, alert webhooks, BI connectors, and jurisdiction-specific dashboards.

Cloud-Native and Event-Driven Scaling Considerations

A production-grade evolution would use cloud-native service boundaries while preserving the current analytic contract. On AWS, a reference implementation could ingest files or messages into Amazon S3, route metadata events through EventBridge, process batch ETL with AWS Glue or containerized jobs on ECS/Fargate, persist curated features in a governed lakehouse, execute model scoring through scheduled workflows, and publish dashboard-ready aggregates through APIs or analytic warehouses. Streaming extensions could use Kinesis or managed Kafka for near-real-time event processing, with schema validation, dead-letter queues, and lineage capture.

Design Option	Technical Tradeoff
Batch-first deployment	Lower operational complexity; appropriate for daily or weekly public aggregate surveillance extracts.
Streaming ingestion	Better timeliness; requires stronger schema governance, idempotency controls, backpressure handling, and monitoring.
Feature store pattern	Improves model reproducibility and reuse; requires metadata management and feature versioning.
API-first dissemination	Supports integration with external dashboards and alerting systems; requires authentication, rate limiting, audit logging, and release management.
Multi-jurisdiction deployment	Supports local configuration and access boundaries; requires tenant isolation, data-use agreements, and jurisdiction-specific calibration.

AI/ML Methodology

Temporal Feature Engineering and Rolling Baseline Design

The feature engineering strategy is organized around state-syndrome time series. For each analytic series, the pipeline computes lagged values, a rolling mean baseline, rolling standard deviation, z-score, seven-period percent change, and trend acceleration. This design gives every observation temporal context and reduces the risk of interpreting normal regional variation as a signal. Rolling baselines are operationally useful because they can adapt to recent local behavior while remaining interpretable to analysts.

Statistical Anomaly Detection

The statistical detection layer uses z-score logic to measure how far an observed visit percentage deviates from its rolling baseline in standard deviation units. Z-scores are transparent, easy to audit, and useful for generating analyst trust because the output maps directly to a familiar baseline-deviation concept. In surveillance settings, this interpretability is important: analysts must understand why a signal was surfaced, whether it reflects volume change, baseline instability, missingness, or a true pattern requiring review.

Isolation Forest Detection

The machine learning layer uses Isolation Forest as an unsupervised anomaly detection method. Isolation Forest is suitable for prototype surveillance analytics because it can identify unusual multivariate patterns without requiring fully labeled outbreak data. The model evaluates a feature vector that includes observed visit percentage, rolling baseline, z-score, percent change, and trend acceleration. This enables detection of patterns that may not be captured by a single threshold, such as sudden increases combined with abnormal acceleration or unusual divergence from expected baseline.

Ensemble Risk Scoring and Analyst Trust

The risk engine combines statistical severity, growth, unsupervised anomaly flags, and persistence into a 0 to 100 score mapped to Low, Moderate, High, and Critical levels. The intent is not autonomous decision-making; the intent is analyst review prioritization. The dashboard exposes model components so reviewers can distinguish a high score caused by sustained growth from one caused primarily by a single unsupervised model flag. This supports explainability, reduces black-box behavior, and creates a pathway for model governance.

Method	Purpose	Operational Tradeoff
Rolling baseline	Recent local expected value for each state-syndrome series	Highly interpretable; sensitive to sparse early windows.
Z-score anomaly	Standardized deviation from rolling baseline	Transparent thresholding; requires stable variance estimates.
Isolation Forest	Unsupervised multivariate anomaly detection	Captures nonlinear unusual patterns; requires calibration and review.
Percent change	Growth relative to lagged observation	Useful for surge detection; unstable when prior values are near zero.
Risk score	Weighted composite of severity, growth, model flag, and persistence	Operationally scannable; weights require validation and governance.

Model Governance, Calibration, and Validation

Operational surveillance models require governance beyond code execution. Thresholds must be calibrated against jurisdiction-specific baseline behavior, seasonality, reporting artifacts, day-of-week effects, facility participation changes, and known public health events. False positives can create analyst burden, while false negatives can delay investigation. A production validation plan should include retrospective event review, analyst adjudication, drift monitoring, performance stratification by geography and syndrome, and documented threshold change control.

- Maintain model cards describing features, thresholds, intended use, limitations, and known failure modes.
- Track model version, feature version, input dataset hash, and scoring timestamp for every generated signal.
- Calibrate thresholds separately for high-volume and low-volume regions to reduce avoidable false positives.
- Use analyst feedback loops to distinguish data quality artifacts from epidemiologically meaningful signals.
- Monitor drift caused by coding changes, care-seeking behavior, reporting interruptions, and facility onboarding changes.

Forecasting Methodology and Operational Limits

Forecasting supports early warning and planning, but it must be presented carefully in public health operations. Short-term syndrome forecasting can help analysts understand expected near-future direction under recent trends, but forecasts should not be treated as authoritative predictions. The prototype documents ARIMA-style forecasting and fallback logic, while the broader architecture allows comparison with Prophet and future sequence models.

Forecasting Method	Use Case	Tradeoff
ARIMA	Interpretable univariate time-series forecasting for short horizons	Works well for simple temporal dynamics; needs stationarity checks and careful parameter selection.
Prophet	Trend and seasonality decomposition with holiday/event extensibility	Useful for operational seasonality; dependency and runtime stability must be managed.
Exponential smoothing	Lightweight fallback for recent trend projection	Low complexity; limited ability to capture complex nonlinear patterns.
LSTM / sequence models	Future deep learning option for multivariate temporal patterns	Higher capacity; requires more data, governance, explainability, and validation effort.

Lightweight forecasting was selected initially because the dataset is historical, aggregate, and intended for prototype demonstration. Simpler methods are easier to inspect, faster to run locally, and less likely to imply unsupported precision. Future deep learning approaches could be evaluated if the platform receives longer time histories, stable data feeds, adjudicated events, and governance controls for model drift and uncertainty communication.

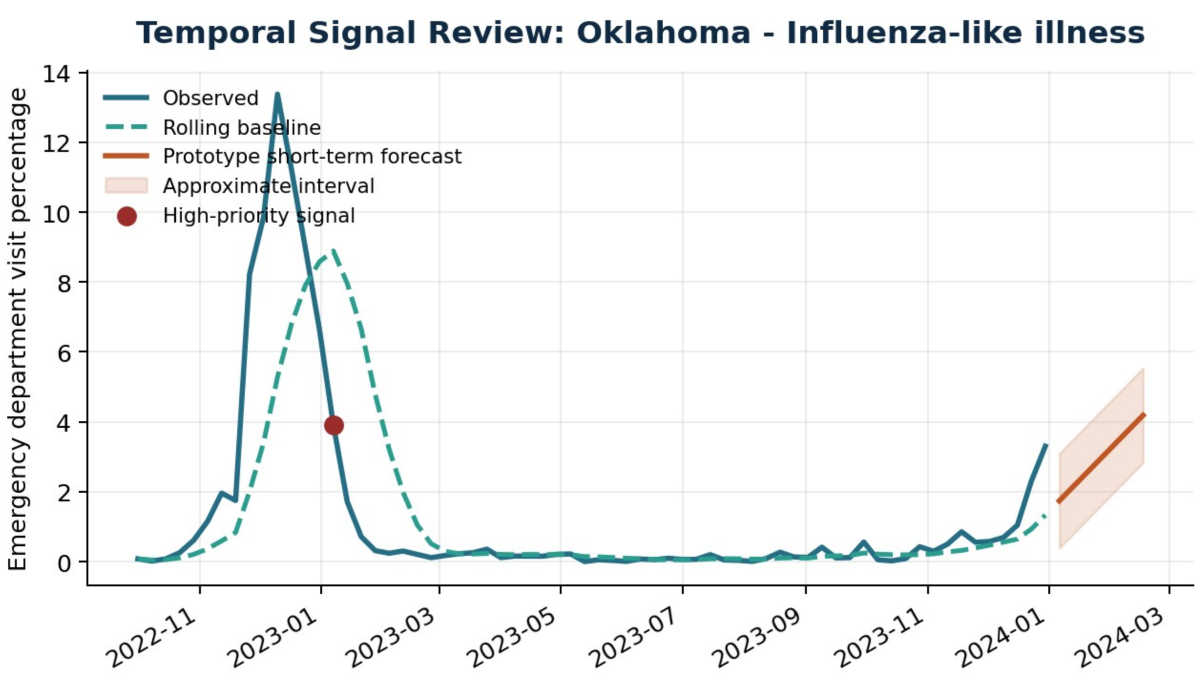

Figure 2. Example observed trend, rolling baseline, and prototype short-term forecast for a high-priority state-syndrome signal.

RAG and NLP Surveillance Assistant

The RAG-based assistant is designed as an analyst workflow augmentation layer rather than a general medical chatbot. Its purpose is to help users ask natural-language questions about current signals, risk rankings, anomaly explanations, data quality limitations, syndrome definitions, and methodology. Retrieval grounding constrains responses to app-generated outputs and curated reference notes, reducing the risk that the assistant fabricates unsupported conclusions.

A production-grade implementation would maintain a vector index over structured surveillance summaries, anomaly tables, data quality reports, model documentation, public health terminology notes, and approved reference material. Queries would be embedded, matched against the governed retrieval corpus, and answered with citations to the retrieved artifacts. The response layer would enforce policy boundaries: no diagnosis, no treatment advice, no claims of official public health direction, and no disclosure of restricted data.

RAG Component	Engineering Role
Retrieval corpus	Model outputs, alert summaries, data quality reports, methodology notes, syndrome definitions, and curated public health references.
Semantic retrieval	Vector search supports plain-language questions that do not exactly match column names or technical labels.
Grounded generation	Responses are generated only from retrieved app artifacts and public reference notes, reducing hallucination risk.
Ontology extension	Future integration with SNOMED CT, ICD-10-CM, LOINC, UMLS, or local syndrome definitions could improve terminology mapping.
Governance controls	Prompt boundaries, source citations, audit logs, restricted corpus access, and refusal behavior for medical advice questions.

Example analyst and policy-maker questions include: Which states show unusual respiratory activity this week? What regions have the highest anomaly scores? Summarize current signals for public health review. Which signals may reflect data quality issues? Explain why a state was flagged as high risk using baseline deviation and model outputs. This pattern supports faster orientation without replacing epidemiological judgment.

Synthetic Data and HL7 Simulation Framework

Synthetic data capability is important for surveillance engineering because many realistic development and validation scenarios involve sensitive healthcare data. The prototype includes synthetic HL7 generation concepts to simulate emergency department message patterns, chief complaints, syndrome categories, timestamps, and geography without exposing PHI. Synthetic streams can be used for pipeline testing, model stress testing, dashboard demonstrations, and training exercises.

GAN-based or other generative simulation methods are most appropriate here as synthetic data generation tools, not as the NLP assistant. A future synthetic module could learn statistical properties of aggregate emergency department trends and generate non-identifiable simulated patterns for outbreak surges, seasonal respiratory waves, reporting delays, facility onboarding changes, and anomalous spikes. Any such module would require privacy review, re-identification risk assessment, utility evaluation, and controls preventing synthetic records from being mistaken for real surveillance events.

Synthetic Capability	Operational Value
HL7 message simulation	Tests ingestion, parsing, schema validation, field normalization, and syndrome grouping logic.
Synthetic trend generation	Creates controlled outbreaks, seasonal baselines, data dropouts, and surge scenarios for model stress testing.
GAN-style simulation	Future option for realistic but non-identifiable aggregate ED trend patterns.
Privacy-preserving analytics	Supports development without exposing PHI, subject to re-identification risk controls.
Validation utility	Allows repeatable benchmark scenarios with known signal timing and severity.

Quantitative Evaluation and Prototype Metrics

The current repository provides measurable prototype outputs from the processed dataset. Some operational metrics, such as true anomaly precision and recall, require adjudicated ground truth labels and cannot be asserted from unlabeled public/synthetic data alone. The evaluation framework below separates observed prototype measurements from production validation metrics that would require labeled events, analyst adjudication, and controlled deployment telemetry.

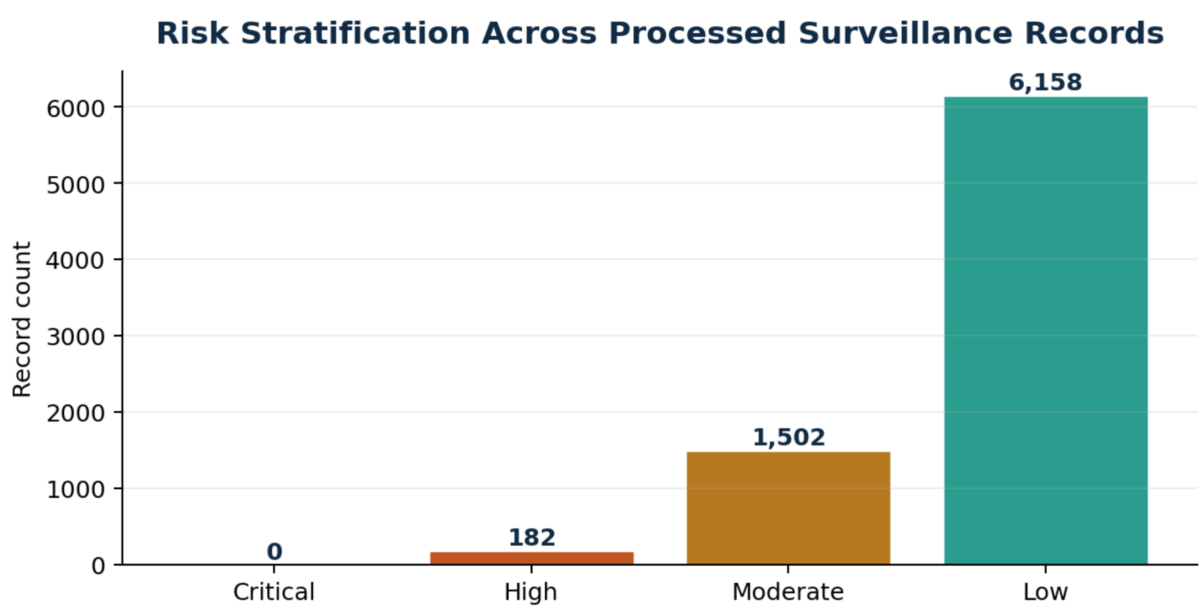

Figure 3. Current risk level distribution across processed analytic records.

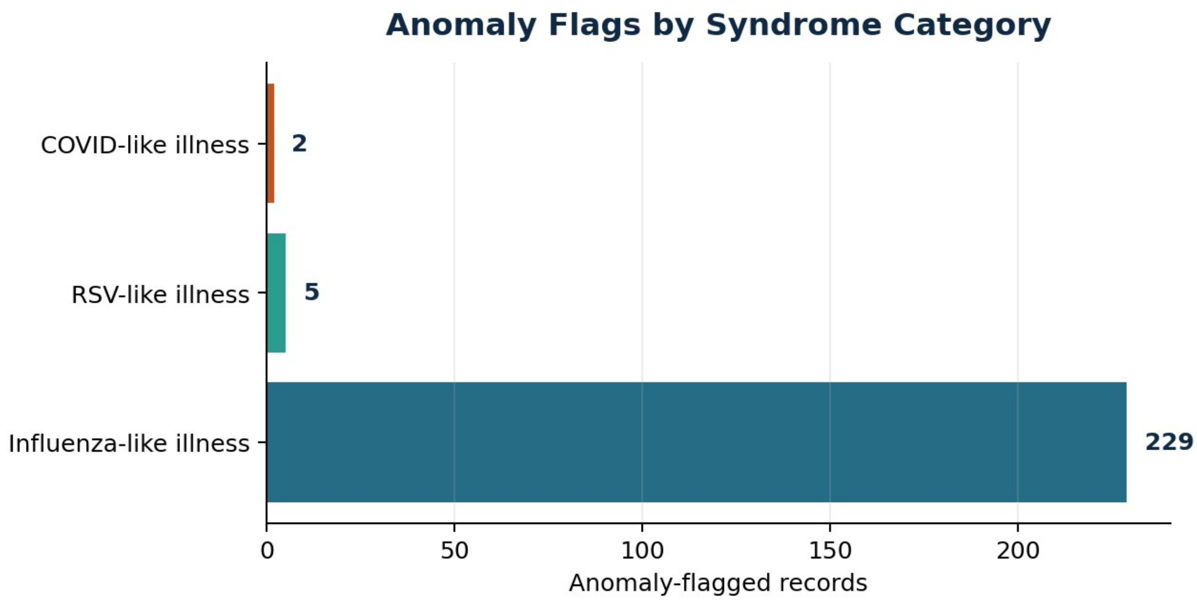

Figure 4. Current anomaly flags by monitored syndrome category.

Metric	Current Value or Validation Status	Interpretation
Data completeness	98.2%	Observed in processed repository dataset.
Duplicate count	0	Observed exact duplicate count after processing.
Anomaly prevalence	236 of 7,842 records	Observed any_anomaly flag count from statistical or Isolation Forest model output.
Model agreement count	0	Observed records where statistical and Isolation Forest anomaly logic both flagged.
High-priority records	182	Observed records mapped to High or Critical risk level.
Average risk score	15.8	Observed across processed records on 0 to 100 risk scale.
Maximum risk score	76.3	Observed highest model-assigned risk score.
Precision / recall	Requires adjudicated labels	Production validation should compare alerts to known events and analyst review outcomes.
Ingestion throughput	Benchmark target: batch-scale thousands to millions of records	Production test should measure rows per minute by deployment environment.
Dashboard response time	Benchmark target: sub-second to low-seconds for filtered aggregate views	Production test should measure p50/p95 latency under concurrent users.

Geospatial Risk Intelligence

Geospatial risk scoring translates record-level model outputs into state or regional prioritization. The current prototype uses aggregate state-level risk summaries and map-ready centroids to show where unusual activity is concentrated. In an operational environment, this layer could be extended to county, health district, hospital referral region, or jurisdiction-specific boundaries, subject to data availability, privacy rules, and aggregation thresholds.

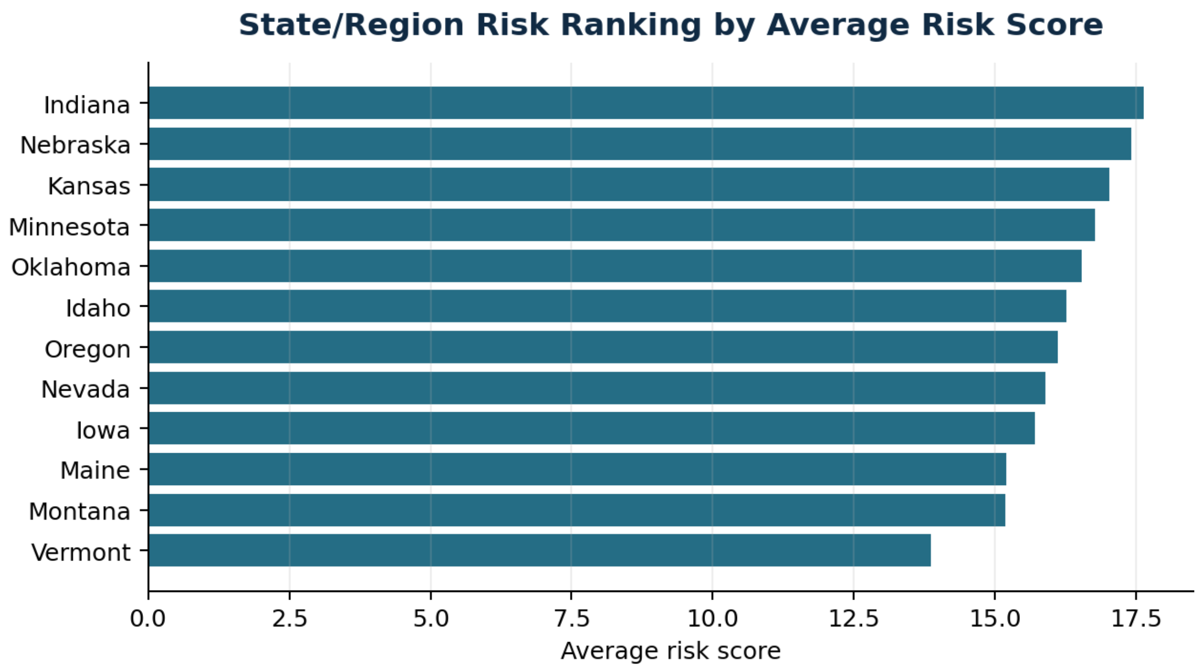

Figure 5. State/region average risk ranking generated from processed model outputs.

State/Region	Avg. Risk	Max Risk	Anomalies	High Signals
Oklahoma	16.5	76.3	9	8
Nebraska	17.4	74.7	9	9
Idaho	16.3	73.2	7	7
Montana	15.2	72.8	9	9
Vermont	13.9	72.6	8	7
Minnesota	16.8	69.5	7	4

Operational Readiness Considerations

A surveillance analytics platform must be evaluated not only for model performance but also for operational readiness. Healthcare and public health systems require strong controls for security, PHI handling, access management, audit logging, data retention, lineage, model governance, and responsible use. The prototype demonstrates the analytic architecture while identifying the additional controls required before any operational deployment.

Readiness Domain	Production Requirement
Cybersecurity	Authentication, authorization, encrypted storage, TLS, dependency scanning, secret management, and network boundary controls.
PHI and privacy	Minimum necessary data, de-identification or aggregation, role-based access, retention policies, and privacy impact review.
Auditability	Input dataset version, model version, feature version, scoring timestamp, user activity logs, and alert disposition history.
Interoperability	Future alignment with HL7 v2, FHIR resources, vocabulary standards, and jurisdiction-specific syndrome definitions.

Responsible AI	Human review, uncertainty communication, no medical advice, prompt boundaries, bias monitoring, and documented limitations.
Continuity	Automated monitoring, retries, dead-letter queues, runbooks, backup strategy, and incident response procedures.

National-Scale Public Health Relevance

Systems with this architecture are relevant to U.S. public health infrastructure because they address a recurring modernization challenge: converting heterogeneous healthcare data into timely, explainable, and operationally usable intelligence. National-scale surveillance requires local flexibility and shared analytic principles. A modular design can allow jurisdictions to adapt syndrome definitions, thresholds, and workflows while preserving common data quality checks, model documentation, and risk communication patterns.

Cross-jurisdiction situational awareness is especially important during respiratory illness surges, emerging infectious disease events, environmental emergencies, mass gatherings, and substance-related harms. A standardized analytic pipeline can help analysts compare trends across regions, identify unusual patterns earlier, and prioritize investigation while maintaining appropriate local control over data and interpretation.

Future Research and Development Directions

Research Direction	Technical Rationale
Federated analytics	Allow jurisdictions to compute comparable metrics locally while sharing aggregate signal summaries.
Privacy-preserving ML	Evaluate differential privacy, secure aggregation, synthetic data validation, and privacy risk scoring.
Adaptive anomaly detection	Tune thresholds dynamically by geography, seasonality, syndrome category, reporting quality, and volume.
Real-time streaming surveillance	Extend batch ETL into event-driven pipelines with schema validation, backpressure handling, and low-latency scoring.
Multimodal intelligence	Integrate structured visits, chief complaint text, lab signals, environmental feeds, and public health reference documents.
NLP summarization	Generate source-grounded analyst briefings with citations, uncertainty statements, and review recommendations.
Ontology-aware retrieval	Map user questions and syndrome language to standardized clinical/public health vocabularies.

Implementation Evidence in the Prototype Repository

Component	Repository Evidence
Data cleaning	src/data_cleaning.py transforms raw surveillance extracts into standardized analytic records.
Feature engineering	src/feature_engineering.py adds lag, rolling baseline, z-score,

	percent change, and acceleration features.
Anomaly detection	src/anomaly_detection.py implements z-score and Isolation Forest anomaly detection.
Risk scoring	src/risk_scoring.py converts model components into interpretable risk scores and levels.
Forecasting	src/forecasting.py contains ARIMA/Prophet/fallback forecasting patterns for short-term trend analysis.
Geospatial analytics	src/geospatial.py provides state/region mapping and risk visualization support.
NLP/RAG assistant	src/rag_assistant.py supports grounded public health search assistant behavior over app data and documentation.
Synthetic data	src/hl7_generator.py provides synthetic HL7-style generation concepts for testing and demonstration.
Application layer	app.py implements the Streamlit dashboard, filters, tabs, metrics, charts, map, RAG assistant, and methodology views.

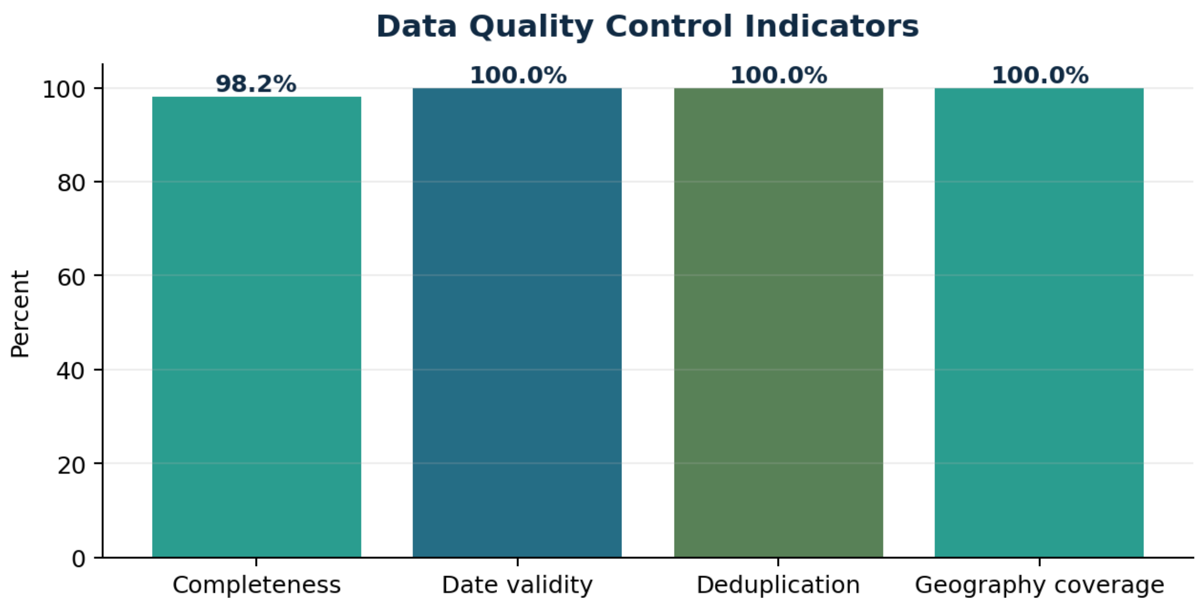

Figure 6. Data quality indicators available from the processed analytic dataset.

Limitations

- The current data boundary is public aggregate and synthetic data, not live operational surveillance feeds.
- The model outputs are prototype analytics intended for analyst review and require validation before operational use.
- Precision, recall, sensitivity, and specificity require adjudicated ground truth labels and cannot be inferred from unlabeled aggregate data alone.
- Forecasting is presented as a short-term analytic pattern, not as authoritative prediction or public health guidance.
- RAG responses must remain grounded in approved app data and documentation and must not provide medical advice or official public health instructions.

- Geographic outputs are state-level in the current prototype and do not represent facility-level or patient-level surveillance.

Official References

U.S. Department of Health and Human Services, Emergency Preparedness and Response: https://www.hhs.gov/programs/emergency-preparedness/index.html

CDC National Syndromic Surveillance Program: https://www.cdc.gov/nssp/

CDC BioSense Platform: https://www.cdc.gov/nssp/php/about/about-nssp-and-the-biosense-platform.html

CDC ESSENCE: https://www.cdc.gov/nssp/php/onboarding-toolkits/essence.html

CDC Data Modernization: https://www.cdc.gov/data-modernization/php/about/index.html

WHO Public Health Surveillance: https://www.who.int/westernpacific/menu/mega-menu/emergencies/surveillance

WHO Epidemic Intelligence from Open Sources: https://www.who.int/initiatives/eios

USA.gov Disasters and Emergencies: https://www.usa.gov/disasters-and-emergencies

Disclaimer

This document describes an independent public health analytics research prototype using public aggregate or synthetic data. It is not an official system of HHS, CDC, WHO, NSSP, BioSense, ESSENCE, or any government agency. It is not intended for clinical decision-making, diagnosis, treatment, or operational public health response.